What the completed runs show

Fixed interim snapshot - the expanded study is not complete. 2026-11-07 21:21 UTC. Current results below are selection validation and fresh confirmation; the new main heldout-100 test has not started.

Verified closed work	Actual amount
New native validation + confirmation episodes	14,980
New main heldout episodes	0 / target 100 per selected configuration and trained model
Closed scientific worker time	3.253 hours; parent clocks counted once
Independent trained models	Seeds 11, 23 and 37; paired rollout starts are not new training seeds

QGF: independently tuned Task 2 choices reached 40/40 fresh confirmation for all three trained models. **Regular D-Flow:** seed 11 raw-Q choices reached 35/40 at lambda 0 and 28/40 at lambda 1. **Stochastic raw-Q D-Flow:** seed 11 reached 40/40 at lambda0 and 37/40 at lambda1. **Original FK:** seed 11 reached 40/40 with its own selected setting. These are not new heldout success rates.

Q averaging: the first completed regular/averaged-Q family has modest changes at overlapping settings: 18 versus 18, 13 versus 15, and 5 versus 5 successes out of 20. Its fresh confirmation and other models remain pending.

Current D-Flow matrix

Condition	lambda	Seed 11	Seed 23	Seed 37
Regular / raw Q	0	confirmed 35/40	pending	pending
Regular / raw Q	1	confirmed 28/40	pending	pending
Stochastic / raw Q	0	confirmed 40/40	pending	pending
Stochastic / raw Q	1	confirmed 37/40	pending	pending
Regular / averaged Q	0	VAL20 complete; CONF pending	pending	pending
Regular / averaged Q	1	TRAIN complete; VAL pending	pending	pending
Stochastic / averaged Q	0	TRAIN complete; VAL pending	pending	pending
Stochastic / averaged Q	1	TRAIN complete; VAL pending	pending	pending

All eight D-Flow conditions use identity surrogate gradients and full-F8 endpoints. Pending means not completed; it does not mean zero success. IPG remains unqualified after engineering failures. No algorithm winner is established.

What the robot and critic do

The current task is **sequential MuJoCo PointMaze navigation**: a simulated point robot moves through walls and corridors toward a goal. Every environment action produces a real next state. These are not synthetic one-point tests or cube-grasp episodes.

Task	Nominal start	Goal	Status in this update
Task 1	(0, 0)	(20, 20)	Full-route/custom-reset QGF probes; separate protocols
Task 2	(0, 20)	(20, 0)	Primary current sampler comparison
Task 5	(20, 4)	(0, 0)	Prior 100-episode results; new expanded-M proposal unrun

Native reset noise, the 1,000-step episode cap and the original success criterion are retained. The point controller changes position and resets velocity each step; this is not inertial legged locomotion. Full simulator/wrapper state is still retained for exact resume.

Learned BC flow and IQL Q/V

The BC flow learns the behavior action distribution from offline data. A goal-specific IQL critic is trained on transition tuples (state, action, reward, next state, termination/mask), with a learned V function for its value targets. Sampling uses the frozen IQL **target critic**, aggregated by the original agent. Actor, Q, target-Q and V checkpoints stay fixed across sampler comparisons.

"Raw Q" means that frozen learned IQL prediction without nearby-action averaging. It does not mean a known true return. "Averaged Q" queries the same learned critic at perturbed actions. The probes add no observed rewards or transitions, and do not train the critic.

Train, validation, confirmation and heldout are separate

Role	Purpose
TRAIN16	Fixed paired data states for prescreening; copies across models are one panel, not independent panels.
VAL20	Compare declared candidates on 20 paired reset/policy contexts.
Fresh confirmation40	Evaluate each literal overall top three on 40 new validation episodes; select using this panel.
Main heldout100	New test only after all 33 guided choices freeze: DF24 + QGF3 + IPG3 + FK3. Currently blocked.

Seeds 11, 23 and 37 identify separately trained models. Seed 23 is not episode 23. Weak results for one model on Task 5 do not establish a broken seed or a critic-only cause.

Methods, coefficients and searched settings

Symbol	Meaning
M	Flow time slices per action: original QGF searches 8/16/32/64. D-Flow endpoints remain full F8.
I	D-Flow latent updates before producing one environment action.
N	IPG/FK particle candidates per state; one action is sampled from the final population.
K	Number of nearby critic-action probes for Q averaging.
kappa / eta	Q-guidance strength / D-Flow latent step size. Equal kappa across samplers does not imply equal accumulated guidance.
lambda / sigma	Latent prior-pull coefficient / Q-neighborhood perturbation scale. Stochastic temperature T is fixed at 1.

D-Flow update actually tested

```
a_i = full_F8(s, z_i); g_i = action-gradient Q_or_Qbar(s, a_i)

z_next = z + eta*(kappa*g - lambda*z)

Stochastic only: add sqrt(2*eta*T)*xi; T = 1
```

The endpoint is integrated through all eight flow steps. The backward endpoint Jacobian is replaced by the identity, so g is a **surrogate**, not the exact full derivative through F8. Regular D-Flow still starts from random flow noise; stochastic D-Flow adds further noise during latent updates.

Original QGF recomputes its clean-action proxy, critic gradient and vector-field update at every flow time slice. Removing the one-step D-Flow endpoint did not remove QGF's original per-slice proxy. IPG/FK use the declared original particle adaptations and a linear-grid time extension for the frozen actor; imputed flow scores and finite-particle corrections remain approximations.

Declared selection domains

DF kappa: 0, .02, .04, .08, .1, .2, .4, .8, 1, 2, 3, 5, 8, 12, 16, 20, 32, 40, 64, 80.
Eta: .001, .003, .005, .01, .03, .1. I: 1, 2, 4, 8, 16, 32, 64, 128, 256.
Averaging sigma: .003, .01, .03, .06, .1, .2, .3, .5; K: 4 or 16. Lambda: separate fixed 0 and 1.

The admitted DF hierarchy is at most 9,240 TRAIN settings/model, not the full 73,440 Cartesian combinations. Seed 11 completed all 9,240. Each family receives its own native shortlist and fresh confirmation. No hidden fourth positive candidate is inserted.

QGF jointly searches M in {8,16,32,64} and 27 kappa values from 0 to 320. Original FK prescreens 720 settings/model: 20 original strengths through 80, M8/16/32/64, N8/16/32 and noise 0/.005/.02. All three FK TRAIN grids are complete.

Fresh confirmation: own selected settings

Seed	Selected method	kappa	Own selected settings	Other	CONF
11	Original QGF	16	M=64	-	40/40
23	Original QGF	320	M=8	-	40/40
37	Original QGF	16	M=16	-	40/40
11	DF regular, raw Q, lambda0	0.1	eta=0.1; l=256	F8	35/40
11	DF regular, raw Q, lambda1	40	eta=0.03; l=256	F8	28/40
11	DF stochastic, raw Q, lambda0	5	eta=0.1; l=128	F8	40/40
11	DF stochastic, raw Q, lambda1	5	eta=0.1; l=128	F8	37/40
11	Original FK	80	M=64; N=32	noise=0.02	40/40

These are independently tuned selections, not matched-kappa experiments and not main heldout results. Raw stochastic DF confirms 40/40 at lambda0 and 37/40 at lambda1, both selecting kappa5, eta .1, l128. Averaged-Q DF confirmation and the other trained models remain pending.

Raw stochastic literal top-three confirmation outcomes: lambda0 = 40/40, 24/40, 22/40; lambda1 = 37/40, 23/40, 10/40. All six fresh40 requests remain in the saved metrics.

Keep the unsuccessful confirmation candidates

Family	kappa	Settings	Fresh40
DF regular/raw L0	0.1	eta 0.1, l 256	35/40
DF regular/raw L0	1	eta 0.1, l 256	31/40
DF regular/raw L0	5	eta 0.1, l 256	21/40
DF regular/raw L1	5	eta 0.1, l 128	24/40
DF regular/raw L1	40	eta 0.03, l 256	28/40
DF regular/raw L1	1	eta 0.1, l 128	0/40
Original FK	80	M64, N32, noise 0.02	40/40
Original FK	8	M8, N16, noise 0	30/40
Original FK	5	M64, N16, noise 0	37/40

For example, regular/raw lambda1 reversed its initial validation order: kappa40 confirmed 28/40, kappa5 24/40 and kappa1 0/40. Saved source/cache reviews found no defect explaining the reversal; it remains observed validation variation. The optional duplicate replay is unrun.

Stochastic raw-Q validation and matched rows

lambda	kappa	eta	l	Success	Role
0	0.1	0.1	128	0/20	new VAL20
0	1	0.1	256	19/20	new VAL20
0	5	0.1	128	20/20	new VAL20
0	64	0.03	256	5/20	new VAL20
0	64	0.001	8	0/20	new VAL20
0	0	0.1	128	0/20	new VAL20
0	BC control	0.03	0	0/20	old BC reference
1	0.1	0.1	256	0/20	new VAL20
1	1	0.1	128	5/20	new VAL20
1	5	0.1	128	19/20	new VAL20
1	64	0.01	128	8/20	new VAL20
1	64	0.001	8	0/20	new VAL20
1	0	0.1	256	0/20	new VAL20
1	BC control	0.03	0	0/20	old BC reference

Existing exact-setting DF overlaps

lambda	kappa	eta	l	Regular	Stochastic
0	1	0.1	256	13/20	19/20
1	1	0.1	128	0/20	5/20
1	5	0.1	128	18/20	19/20

These are all literal raw-DF mode-to-mode overlaps in the four completed seed-11 VAL20 lists. Every paired initial state and reset/policy ID matches. Common canonical BC controls are excluded from the mode comparison and separately accounted for.

Interpretation: these rows do not establish a deterministic advantage. They also do not establish a general stochastic advantage: this is one trained model and a posthoc overlap of small validation panels. The separately planned matched heldout curves have not run.

One canonical BC B20 is referenced by four raw-family lists (three cache references). References are not new executions or independent model seeds. Kappa0 with latent prior/noise is not generally canonical BC.

All 22 averaged-Q validation requests

Task 2 / seed 11 / regular identity-Jacobian full-F8 D-Flow / fixed lambda0. All selection-validation results are retained, including zeros. The displayed 440 records comprise 420 new episodes plus 20 references to the existing canonical BC batch.

#	kappa	eta	I	sigma	K	Success
1	0.1	0.1	256	0.003	4	18/20
2	1	0.1	256	0.003	16	15/20
3	5	0.1	256	0.003	16	5/20
4	64	0.03	256	0.003	16	4/20
5	64	0.03	256	0.003	4	4/20
6	64	0.03	256	0.01	4	4/20
7	64	0.03	256	0.01	16	4/20
8	64	0.03	256	0.03	4	4/20
9	64	0.03	256	0.03	16	4/20
10	64	0.03	256	0.06	4	4/20
11	64	0.03	256	0.06	16	4/20
12	64	0.03	256	0.1	4	4/20
13	64	0.03	256	0.1	16	5/20
14	64	0.03	256	0.2	4	4/20
15	20	0.1	256	0.2	16	5/20
16	20	0.1	256	0.3	4	3/20
17	20	0.1	256	0.3	16	4/20
18	20	0.1	256	0.5	4	3/20
19	20	0.1	256	0.5	16	4/20
20	3	0.03	1	0.03	4	0/20
21	0	0.001	1	0.5	16	0/20
22	BC	0.03	0	0	4	0/20

These are the literal hierarchy-selected candidates, not a full Cartesian kappa/eta/I/sigma/K sweep. Kappa changes for some large-sigma rows, so the full table must not be labelled a fixed-kappa sigma-only curve. More K is not guaranteed better for a particular finite bank.

Same kappa / eta / I	Raw Q	Averaged Q
.1 / .1 / 256	18/20	18/20 (sigma .003, K4)
1 / .1 / 256	13/20	15/20 (sigma .003, K16)
5 / .1 / 256	5/20	5/20 (sigma .003, K16)

The strongest averaged-Q VAL20 candidate uses a very small sigma (.003). Fresh top-three confirmation has not yet completed. This panel alone does not show that averaging fixes unseen-action Q or improves the underlying critic.

What Q averaging and lambda1 mean

Q averaging asks for a neighborhood score. Hold the state fixed, perturb candidate action a K times, clip the neighbors to the action range, read the same learned critic at those neighbors, and average the K predictions.

```
Qbar(s,a) = mean_j Q(s, clip(a + sigma*epsilon_j))
```

Illustration only: Q predicts 10 at action .50. Four neighboring probes predict 2, 3, 3 and 2; their average is 2.5. This deliberately changes the score being optimized. It does not establish that the true return of exact action .50 is 2.5, and no new reward or transition was observed.

Sigma controls the perturbation scale; K controls approximation size. Our finite antithetic bank stays fixed during each action generation. K4 and K16 banks are not guaranteed nested. Regular and stochastic DF can both use averaging; per-update latent noise is a separate operation.

Averaging may weaken a narrow false peak, leave a broad wrong region wrong, or blur a genuinely narrow good action. When nearby predictions are already similar, its effect can be small. A gain may reflect a changed smoothing objective rather than a better sampler or critic.

Lambda1 is the setting lambda = 1

The term -lambda*z pulls the latent/noise input toward zero. With z=2, eta=.1 and no Q/noise contribution, lambda0 leaves z=2 and lambda1 produces z=1.8. It acts in latent space, not directly on the final action or a measured data-support distance. Repeated deterministic shrinkage can reduce diversity and hurt. It adds no missing-Q information.

Primary literature: established smoothing ideas

Nachum et al., ICML 2018: Equation 9 defines Gaussian-averaged action values; Smoothie learns a smoothed critic and Gaussian policy. Primary paper.

Fujimoto et al., ICML 2018 (TD3): Section 5.3 smooths critic training targets using noisy nearby actions. Our frozen-IQL inference-time averaging is a different application. Primary paper.

Failures, unseen-action probes and limits

Study / fix	Actual outcome; retained limitation
Original expanded IPG/FK gate	33 IPG absolute solve-residual failures; 15 learned IPG cases missing. Original FK-only factorized completion passed; IPG did not.
IPG saved algebra	5,760 recorded systems diagnosed precision sensitivity. This is local matrix evidence, not policy qualification.
IPG mixed precision V2	Failed strict abstract dtype guard before sampler cases. Separate abstract capture mapped 28 non-solver float64 nodes.
IPG V3 nested contexts	One abstract proof passed; first production API entry failed MLIR int64-to-int32 lowering. Zero returned sampler results, reference calls or completed cases.
IPG V4 explicit helpers	Completed 144 synthetic cases: 134 PASS / 10 FAIL. All 12 strict dtype proofs passed, with zero float64 outside the solve. 288 production/reference calls; no learned/native cases. No qualification, C release or automatic retry.
Other engineering limits	Old I512 gate FAIL and 25 raw-FD32 failures retained. Later qualified FD64/AD checks are factorized, not exhaustive domain proofs.

Did QGF fail on unseen actions?

The standard full-route FULL/HOLE QGF scan completed 3,240 VAL episodes but had zero joint withheld state-action exposure. It is not a direct unseen-action-Q stress result. A custom right-start probe did expose the withheld joint region: 2,076 exposed episode executions / 6,533 steps. Yet each of six arm/model settings had a positive-kappa best 20/20 VAL. This is custom-reset performance, not official Task 1 reset performance or proof that Q is correct.

The larger hard-Q proposal remains unadopted. Independent review found that its proposed state panel missed a direct hole anchor and assumed incorrectly nested K banks. A separate small TRAIN-only score/gradient characterization has passed source review, with an actual hole-state anchor; it has not run and cannot establish true Q error.

Future discriminating cases should separate narrow false peaks, broad wrong regions and legitimately narrow good actions. Fixed-policy rollout values must name their continuation policy; learned-V Bellman residuals are only self-consistency. No QGF failure or D-Flow win is required.

Counts, reproducibility and remaining work

The closed ledger at snapshot has **14,980** new native VAL/CONF episode executions and **8,547,396** active external action rows. Active-at-snapshot attempts and their in-progress increments are listed in the machine-readable snapshot; they are additional, not zeros. Action rows are not batch sampler API calls.

Count boundary	Treatment
Initial Task2 QGF M8	1,620 executed VAL episodes referenced once inside the joint M scan.
New QGF M16/32/64	4,860 additional VAL; fresh top-three confirmation 360.
Latest two-family DF VAL batch	580 record views = 540 new episodes + 40 views of one old BC B20. Outer 3,936.508402 s counted once.
Raw stochastic DF fresh confirmation	240 new CONF episodes, 134,307 action rows. Outer 487.299440 s counted once.
Latest original FK seed11	140 new VAL + 120 new CONF episodes. Prior TRAIN prescreens contain no native episodes.
Historical Task5	Prior 42 x 100 = 4,200 heldout episodes belong to an older frozen protocol; not new expanded-M tests. Smaller40 branch seeds23/37 unrun, not zero.

Historical Task5 reference only: QGF seeds11/23/37 = 96/100, 24/100, 100/100; original IPG = 98/100, 65/100, 100/100; FK = 100/100, 54/100, 100/100. Different task, settings and cohorts prevent treating these as the current Task2 comparison. Expanded Task5 M-selection remains a source proposal.

Bound evidence

The adjacent SNAPSHOT.json binds all input paths and full SHA256 values, frozen selections, exact configs and report source. PLAN_SNAPSHOT.json preserves the closed/active ledger. MODEL_REFERENCES.json carries actor/critic checkpoint and component-tree references; checkpoints and large datasets are referenced rather than copied. This is a progress snapshot, not the final reproducible research bundle.

Sampler a615c61d284c; DF native design eb2fe2102396; same-phase batching b40a2d23d82b; original FK c1f7c500f511; latest DF audit 6c3768522655; FK confirmation audit feeb17c45a9e. Full digests are in JSON.

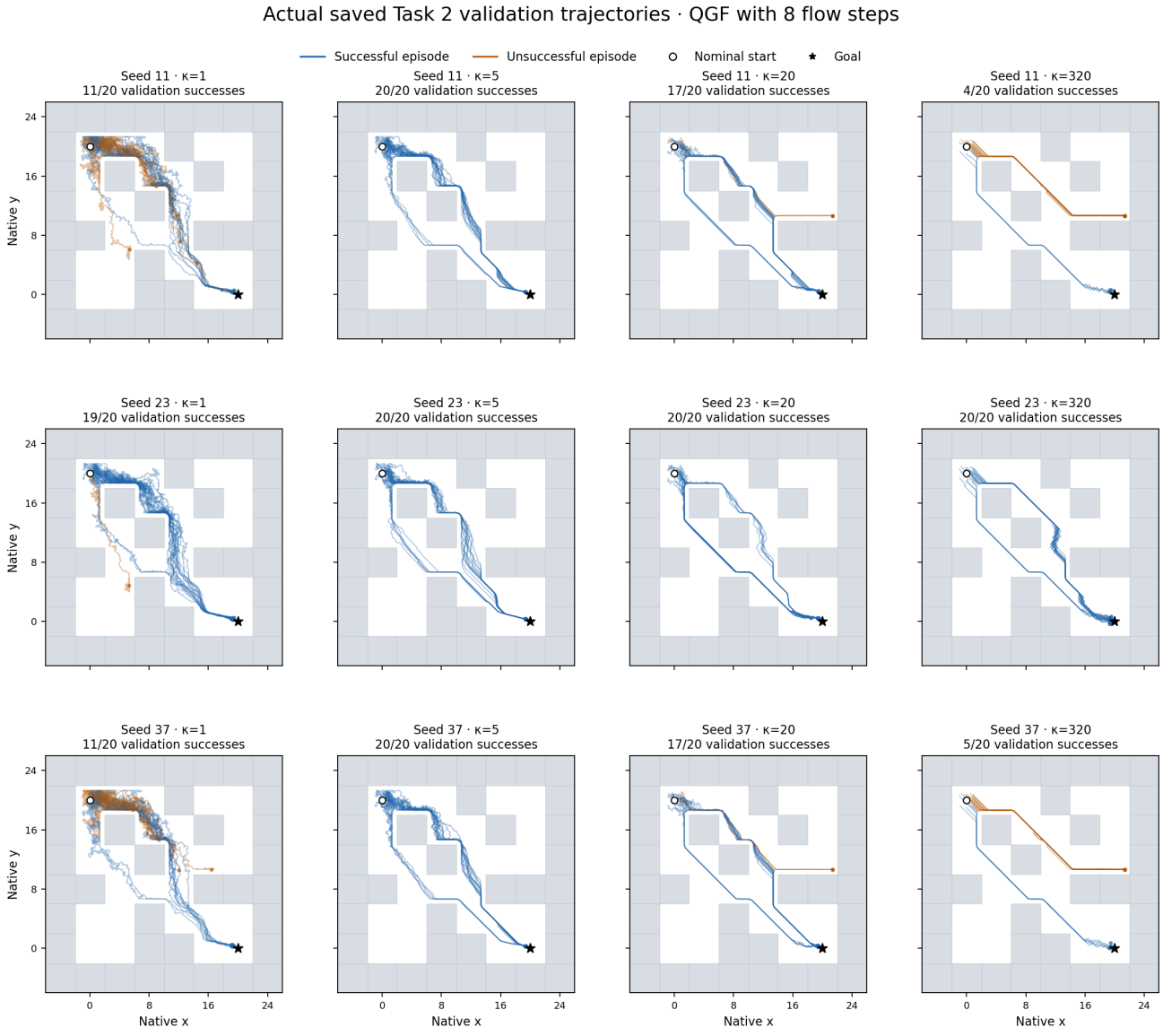
All bound source/metric/audit files are rehashed before the snapshot closes. Old delivered reports and archives are not rewritten. This snapshot is not a fresh exhaustive audit of every historical archive; final closure still requires archive, episode, selection, source and bundle verification.

Next work

Complete the remaining independent DF/FK/IPG validation choices without heldout tuning; qualify or honestly retain failed IPG branches; freeze all 33 guided choices before new main100; run the separately declared matched curves and harder characterization only after their own source/cohort/budget releases. No silent reduction in candidates, iterations, episodes or slow methods. The next progress and final reports include PDF; this interim report is not completion.

Saved Task 2 trajectories: context only

Previously completed original-QGF M8 selection-validation scan. Four kappa values are shown for each trained seed; all 20 saved episodes in each panel are plotted. This 240-record subset is included in the earlier 1,620 executions and adds zero trials. It is not a replay, heldout panel or the new joint-M selected result.



Posthoc illustration: four κ values chosen from the completed 27-value validation scan.
All 20 saved episodes shown in each panel; 240 represented episodes, zero new trials. This is not heldout evaluation or a replay.

Blue paths succeeded, orange paths did not; circles mark nominal starts and stars mark the goal. This shows the physical navigation problem and why both under- and over-guidance can matter in one trained model. The plot is an illustration of saved validation behavior, not a controlled causal claim.